INCH-POUND

MIL-DTL-17/177B
19 June 2015
SUPERSEDING
MIL-C-17/177A
22 March 1988

MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, TRIAXIAL, 95 OHMS, M17/177-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

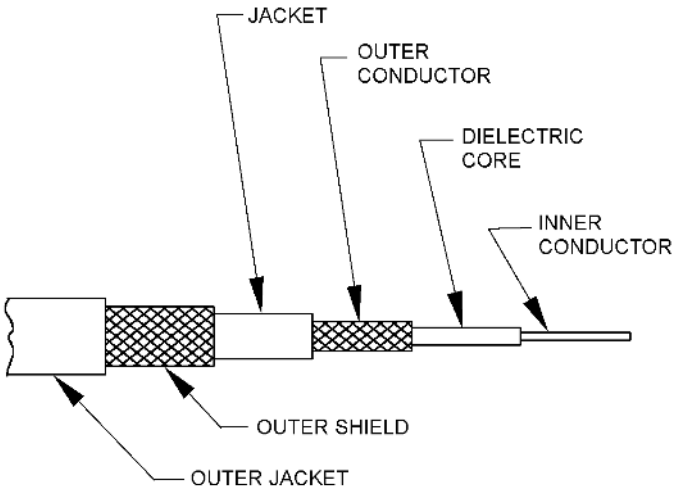


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of silver-coated, annealed-copper-covered, steel wire, each strand .004 inch diameter. Overall diameter: .012 ± .001 inch.
Dielectric core	Type F-1: Solid extruded PTFE. Diameter: .102 inch ± .003 inch.
Outer conductor	Single braid of #38 AWG silver-coated copper wire. Diameter: .124 inch maximum. Coverage: 91.0% nominal Carriers: 16 Ends: 7 Picks/inch: 12.0 ± 10%
Inner jacket	Type IX: Clear transparent FEP. Diameter: .141 ± .004
Outer shield	Single braid of #38 AWG silver-coated copper wire. Diameter: .163 inch maximum Coverage: 91% nominal Braid angle: 38 nominal Carriers: 16 Ends: 8 Picks/inch: 14 ± 10%
Outer jacket	Type IX: FEP Diameter: .184 ± .005

AMSC N/A FSC 6145

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ENGINEERING INFORMATION:

Continuous working voltage: 1,100 V rms, maximum.
Operating frequency: 3 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 10 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.
Engineering note: This cable is useful in shield critical, high temperature applications (see connector series “TRB” and “TRT” in accordance with MIL-PRF-49142.)

REQUIREMENTS:

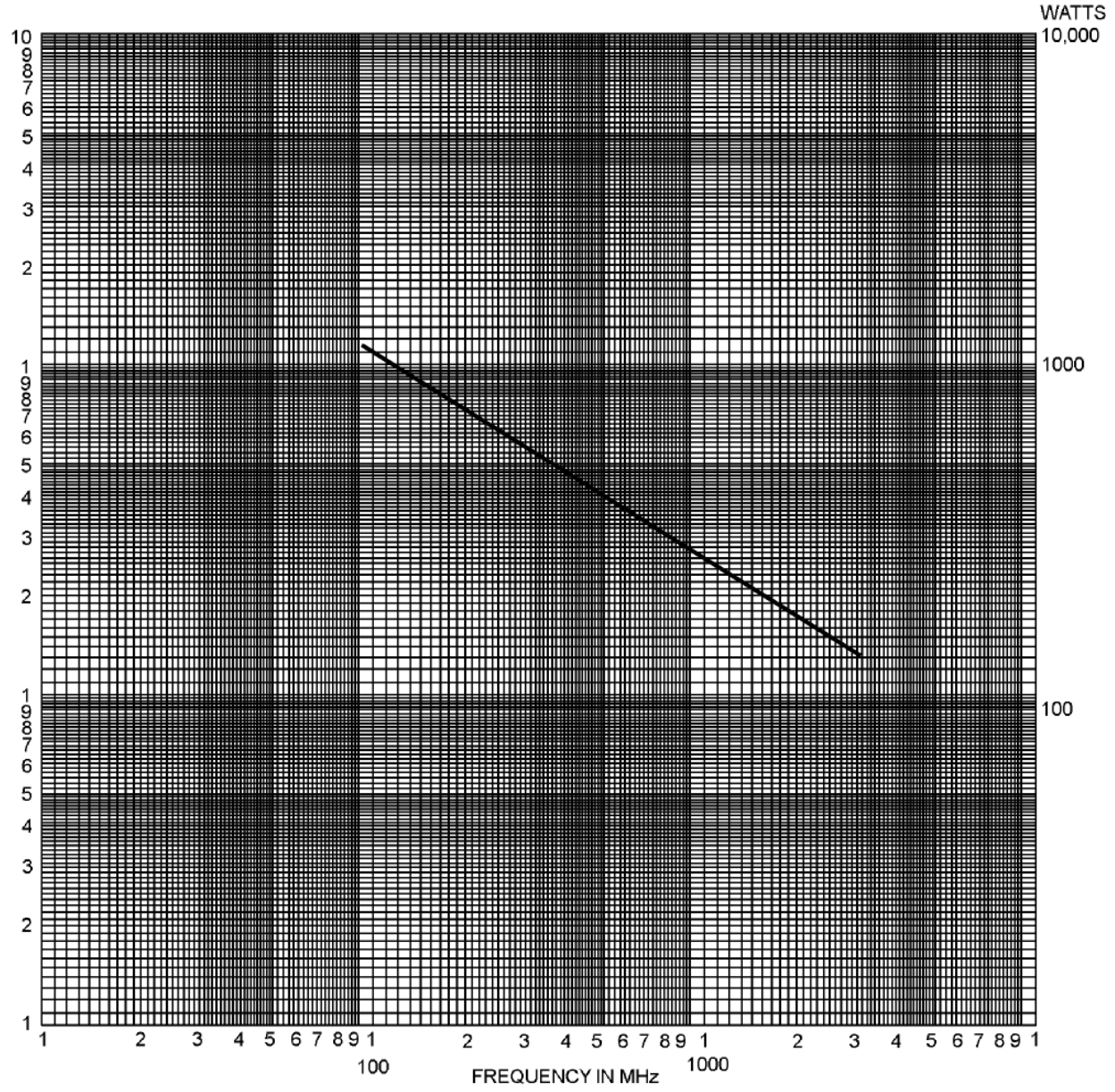
Dimensions, configuration, and descriptions: See figure 1 and table I.
Environmental and mechanical:
Visual and mechanical examination:
Out-of-roundness: Not applicable.
Eccentricity: 10 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 1.5 pounds, minimum; 4 pounds, maximum.
Aging stability: Not applicable.
Stress-crack resistance: +230°C ± 5°C.
Outer conductor integrity: Not applicable.
Cold bend: -55°C ± 2°C.
Dimensional stability: +200°C ± 5°C.
Inner conductor from core: 0.187 inch, maximum.
Inner conductor from jacket: 0.250 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Applicable.
Weight: 34 pounds per 1,000 feet, maximum.

ELECTRICAL:

Continuity: Applicable.
Spark test: 2,000 Vrms, +10, -0 percent.
(2,000 V rms +10, -0 percent applied between the inner and outer conductor with the outer conductor grounded; 200 V dc minimum applied between the outer conductor and the outer shield with the outer shield grounded).

Insulation resistance: Not applicable.
Corona extinction voltage: 1,500 V rms, minimum.
Characteristic impedance: 95 ohms ± 5.
Attenuation: 17 dB per 100 feet, maximum at 400 MHz.
Structural return loss: Not applicable.
Capacitance: 17.4 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Phase stability: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

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Maximum power at 25°C, sea level ——————

FIGURE 2. Power rating.

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Part or Identifying Number (PIN): M17/177-00001.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-49142

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2015-029)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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